ADVISOR BRIEFING · STUDIES RUN 1–12 JULY 2026 · SOLAR-CHAIN BRANCH

Does modeling weather persistence improve optimal solar-UAV dispatch?

Five studies benchmarking dynamic-programming dispatch policies — solved under IID vs Markov-chain weather models — against a tuned threshold heuristic, the baseline a practitioner would actually deploy. All arms run on identical paired historical weather episodes.

+3.7
Wind-chain DP above the threshold baseline (pooled Δreward, 660-cell grid) — the only policy above it

−11.4
IID-solved DP *below* the threshold baseline — significantly worse in 637/660 cells

≈ 0
What solar persistence adds: the solar arm tracks IID (−11.7 vs threshold) at every bin resolution

pen ≳ 150
Regime where the threshold beats even the wind-chain DP (−9.3 at penalty 640)

Bottom line

Against the deployable baseline, the DP policy only earns its complexity when it models wind persistence. The IID-solved DP sits −11.4 below the tuned threshold heuristic (worse in 637/660 cells); adding a wind Markov chain swings the same solver to +3.7 above it — a +15.1 swing, significant in 631/660 cells and never significantly negative. Solar persistence contributes nothing at hourly data resolution: the solar arm is indistinguishable from IID. The wind-chain margin over the threshold survives a native-hourly robustness check, widens with mission duration (+2.3 at 30 days → +8.3 at one year), and has a known limit: beyond failure penalty ≈ 150 the threshold overtakes every DP arm.

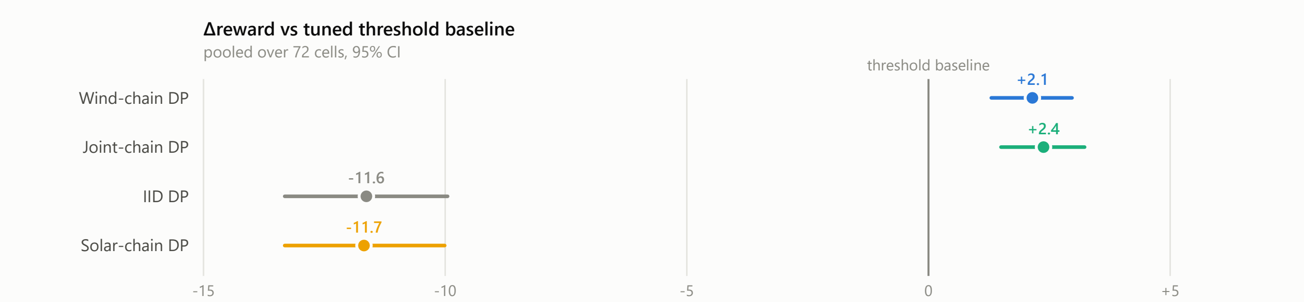

Fig 1 — Main ablation: every DP arm vs the tuned threshold baseline (72 cells). Pooled paired Δreward per episode, 95% bootstrap CIs over cells; 4 sites × 3 capacities × 3 penalties × 2 seasons, 60-day missions, 3 000 paired episodes/cell. Only the wind-informed arms clear the baseline (wind +2.1, joint +2.4); the IID and solar-only arms sit ≈ −11.6 below it. Joint (wind+solar) is indistinguishable from wind alone, and a follow-up sweep of solar bin counts (2–8) found no resolution at which solar helps.

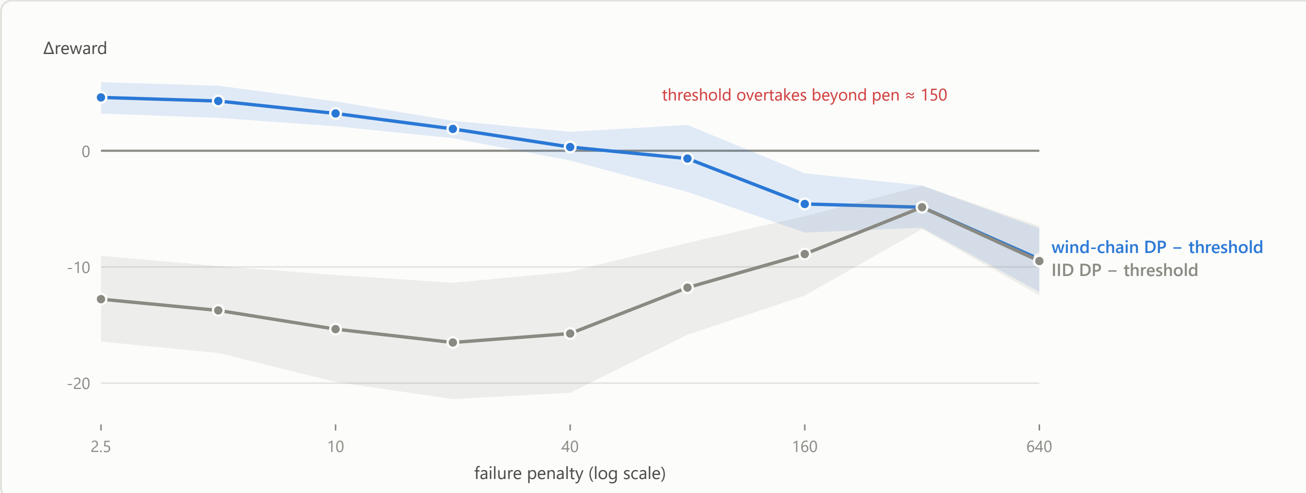

Fig 2 — Both DP arms vs the threshold baseline across failure penalty (300 Wh, 5 sites × 2 seasons, 90 cells). The IID arm is below the baseline at every penalty. The wind-chain arm is above it at low-to-mid penalties (+4.6 at 2.5) but crosses below near penalty ≈ 150 — at extreme risk-aversion both DP arms converge to near-never-fly, persistence information becomes worthless, and the threshold's lower failure rate (4.7% vs 6.1%) wins. The DP advantage is a low-to-mid-penalty phenomenon.

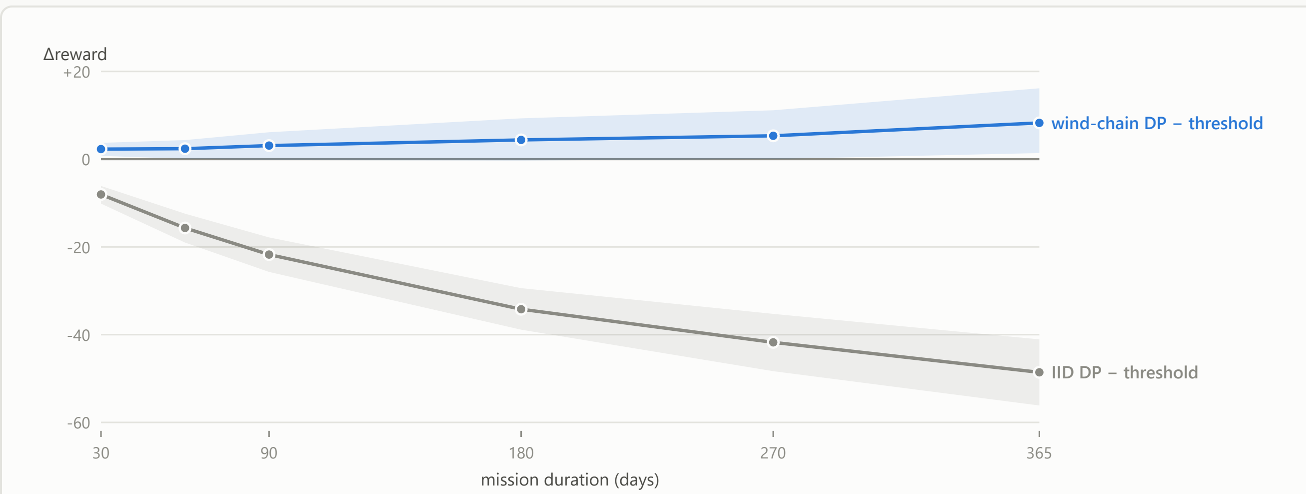

Fig 3 — Both DP arms vs the threshold baseline across mission duration (300 Wh, 5 sites × 3 penalties, 90 cells). The wind chain's margin over the threshold widens with mission length (+2.3 at 30 days → +8.3 at one year), while the IID arm falls ever further below the baseline (−8 → −49). Longer autonomy makes persistence modeling more valuable, not less.

What was run, and why

STUDY	QUESTION	OUTCOME
Bin validation Jul 3	How should the wind chain discretize wind speed — and how many bins?	**Equal-occupancy quantile bins win;** gains saturate by ~8–12 bins. Placing bins in failure space (user hypothesis) refuted — it collapses resolution into the calm regime and the policy never flies.
Markov ablation Jul 6	Which persistence channel matters: wind, solar, or both? (5 arms, 72 cells)	**Wind +13.8 vs IID** (sig. in 68/72 cells); solar ≈ 0 and harmful at Hawaii; joint ≈ wind. Only chain arms beat the tuned threshold (+2.1); IID loses to it (−11.6, 71/72 cells).

Solar resolution Jul 7	Is the solar null just a too-coarse 2-bin artifact?	**No.** Null at every bin count 2–8 (all CIs span 0); the Hawaii degradation persists and worsens with more bins, so it is real, not a binning artifact.
Timestep check Jul 11	Is the wind benefit an artifact of interpolating hourly weather to 15 min?	**Survives on native hourly data** (dt = 60): still significantly positive everywhere at 55% of the per-step magnitude (+0.0013 vs +0.0024 reward/step). The finding is real; its size partly rides on sub-hourly decision granularity.
Thesis sweep + penalty extension Jul 11–12	Full response surfaces (11 capacities × 6 penalties × 5 sites; durations to 1 yr) and the high-penalty limit (to 640).	**Wind +15.1 pooled** (631/660 cells); benefit and absolute reward peak at 250–300 Wh; benefit peaks at penalty ≈ 10 and grows with duration. Surprise: **threshold overtakes DP beyond penalty ≈ 150** — the predicted "threshold collapse" inverted.

Caveats (known, scoped)

Solar claim is scoped to hourly source data — the reanalysis record has no sub-hourly cloud variability, so "solar persistence is useless" is untested at finer resolution. **dt-60 retention is confounded** between interpolation-inflated persistence and decision granularity. All results use one common-random-number seed set and 7-day bootstrap blocks; seed/block-length robustness is on the follow-up list, as is the 5-bin wind-chain setting (a chosen default, not an optimum).

Metric: mean paired per-episode Δ(total mission reward), 95% CIs — percentile bootstrap over cells (10 000 resamples) pooling per-cell paired deltas; per-cell CIs are paired-episode bootstraps (CRN verified). Reproduction: `docs/markov_ablation_experiment.md` (design + decision log) · cells CSVs under `results/{markov_ablation, thesis_sweep, penalty_ext, markov_dt60, markov_solar_res}/_analysis/` · scripts `SolarSimulator/Scripts/{run_chain_sweep, compare_markov_ablation, penalty_curve_analysis}.py`.